

PETROSHIELD AI

National Energy Security Command Center

A Fully Implemented, Data-Grounded Multi-Agent Digital Twin

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Our Real Engineering Upgrades & Implementation:

- Semantic Ingestion: GDELT & RSS Scraping every 8 seconds.
- Grounded Price Data: Live Daily spot prices fetched from US EIA API.
- Optimization Solver: SciPy-powered LP alternate supplier rerouting.
- Real-Time Caverns: Schedules Padur & Mangaluru releases to keep run rate above 70%.

Live URL: <http://localhost:3000> | Backend: <http://localhost:8000>



India's Crude Import Exposure

- 85%+ Crude Import Dependency: India imports over 4.5 million barrels per day, sustaining national growth.
- Critical Chokepoint Constraints:
 - Strait of Hormuz: Carries ~60% of Persian Gulf imports.
 - Red Sea / Bab-el-Mandeb: Regional conflict forces tankers around Cape of Good Hope (+12-15 transit days).
- The Engineering Gap in Existing Systems:
 - Siloed spreadsheets use static templates & mock data.
 - They fail to map text context, cannot resolve crude grade compatibility, & cannot calculate cavern runways.
- Our Geospatial Evidence Depth:
 - Resolves raw news into latitude/longitude coordinates.
 - Automatically maps blocked corridors and routes on the Leaflet GIS digital twin map interface.

Visualizing Global Chokepoints

[Chokepoint Exposure Map]

PetroShield AI: Pre-emptive, Data-Grounded Decision Command Center

Pillar 1: Graph-RAG Infrastructure

We modeled India's entire import network (suppliers, corridors, storage, refiners) as a directed semantic Graph. News entities are resolved and mapped to specific graph nodes dynamically in real-time.

Pillar 2: Live Simulation Engines

Spot prices are fetched live from the US EIA API. Volatility parameters are calibrated on real FRED daily returns. Pricing curves are generated via 5,000 path Monte Carlo (GBM) dynamically.

Pillar 3: Executable Mitigation Solver

SciPy LP optimizer solves procurement rerouting matrices. Strategic cavern releases (Padur/Mangaluru) are scheduled to keep refineries above their 70% minimum operating threshold.

Our Implemented Multi-Agent Architecture (The 5 Subagents)

Subagent Name	Core Task & Library Used	Implemented Role in Security Pipeline
Agent 1: Risk Intelligence	Entity Resolution & Bayesian Scoring (NetworkX)	Fuses live feeds, EIA prices, and Graph topology to score risk.
Agent 2: Scenario Modeller	5,000-path Monte Carlo price forecasting (SciPy)	Calibrates price volatility on FRED historical Brent crude data.
Agent 3: Procurement Agent	Linear Programming (LP) optimization (SciPy)	Reroutes supply and reallocates imports under shipping wait times.
Agent 4: SPR Advisor	Inventory drawdown & Refills (Graph Traversal)	Schedules cavern releases to keep refineries above 70% run rates.
Agent 5: Executive Briefing	Markdown Synthesis (FastAPI)	Compiles decision briefs and commands for the Cabinet Secretary.
Agent 6: Gemini Auditor	Supervisory validation logic (Gemini API)	Audits all calculations and signs off briefs (Supported by key rotation).

Operational Action Loop & Data Flow Pipeline



EVALUATION PERFORMANCE BENCHMARKS (END-TO-END)

8-Second Lead Time	< 200ms Early Stop	< 2.5s Execution	Multi-Key Rotation
Queries live RSS/GDELT feeds continuously in the background.	Low-risk signals are gated immediately to prevent downstream pipeline	Runs full 6-agent pipeline from raw news signal to Cabinet keying	Rotates active Gemini keys to completely eliminate 429 errors.

Our Core Computational Features

Pydantic Event Extractor (Agent 1)

- Replaced keywords with Gemini JSON schema extraction.
- Extracts type, dates, severity, & affected corridors.
- Resolves raw text entities dynamically to Graph IDs.
- Built a robust local keyword parser fallback.

Grounded Monte Carlo Forecaster (Agent 2)

- Projects 5,000-path pricing paths using GBM.
- Uses real spot prices from the US EIA API.
- Volatility parameter calibrated on historical FRED data.
- Produces explicit, testable scenario assumptions.

Adaptive LP Procurement Rerouter (Agent 3)

- SciPy LP optimization solver replaces static lists.
- Minimizes cost, transit times, & route risk.
- Restricts allocation to active suppliers (ADNOC/US).
- Checks API gravity & sulfur grade compatibility.

Active Cavern Planner (Agent 4)

- Schedules releasing caverns (Padur/Mangaluru/Vizag).
- Keeps refiners above 70% min operating thresholds.
- Sets replenishment triggers once price drops below ceiling:

$$\text{Buy Price Ceiling} = \text{Peak Price} * 0.90$$

Why PetroShield AI is the Industry Standard

Dimension	Traditional Excel-Based Systems	Our PetroShield AI Implementation
Realism	Mock Data: Fictional templates and static reports.	Live Grounded Data: Direct EIA API + live RSS streams.
Parsing	Keyword Match: Scanning misses negations & context.	Gemini Pydantic: Deep semantic context extraction.
Decisions	Siloed Worksheets: Logistics and storage separated.	SciPy LP Solver: Fuses cost, grades & wait times in one LP.
Robustness	Hangs on Failure: API rate limits crash the pipeline.	Key Rotation: Alternates keys; features local fallbacks.
Safety	Unverified: Cabinet signs briefs without verification.	Double-Audited: Gemini AI validates physical cavern limits.

Core Differentiation Value:

By embedding mathematical agents (Agent 1-5) underneath an Generative AI Auditor (Agent 6), PetroShield resolves the classic 'Hallucination Problem' of LLMs while preserving full natural language explainability for policymakers.

Platform Technologies Used in Our Implementation

Backend (Python)

- FastAPI: High-speed REST/WS routers.
- NetworkX: Semantic Directed Graph DB.
- SciPy & NumPy: LP optimizer solvers.
- Uvicorn: Async server process gateway.

Frontend (TypeScript)

- Next.js 16: Server-rendered dashboard.
- Leaflet.js: Interactive GIS mapping.
- Recharts: Monte Carlo price graphics.
- WebSockets: Live log/alert terminals.

AI & Data Layer

- Google Gemini API: Audit & narrative briefs.
- Multi-Key Rotation: 429 quota guard.
- EIA API v2: Spot pricing stream.
- AISStream WebSockets: Telemetry feeds.

Expansion Roadmap for PetroShield AI Platform

Phase 1: Dark Tanker Vessel Tracking (Satellite SAR)

Integration of synthetic-aperture radar (SAR) satellite imagery to detect and track 'dark vessels' that switch off their AIS transponders near critical chokepoints like the Strait of Hormuz.

Phase 2: Automated Sanctions Verification

Direct connection to the OFAC and global compliance databases, automatically blocking imports or shipping charters involving sanctioned shipping registries or blacklisted corporate owners.

Phase 3: Multi-Commodity sovereign scaling

Expanding the NetworkX Graph database and SciPy LP optimizers to track and reroute other critical national imports, including LNG pipelines, coal maritime lanes, and EV battery minerals.

Commercial Feasibility & Platform Scaling Models

Sovereign Command Center

- Sold as custom tactical dashboards for oil-importing governments.
- Integrates direct state SCADA inventories and reserve cavern drawdown controllers securely.

Enterprise B2B SaaS

- Licensable to private refinery giants (Reliance, Nayara, Rosneft).
- Optimizes commercial shipping logistics and manages fuel hedging risk portfolios dynamically.

Cloud-Scale Simulations

- Containerized deployment using Kubernetes clusters.
- Scales Monte Carlo models to 1,000,000 parallel path simulations for global commodity trading desks.